

Emotion-Aware Interfaces: Bridging Human and Intelligent Machines

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Abstract

Emotions are fundamental to human communication and play a critical role in shaping interactions. As Artificial Intelligence advances, there is a growing demand for systems that can not only process information but also understand and respond to human emotions. Emotion-aware interfaces represent an essential step toward this goal, enabling machines to recognize users' affective states and adapt their responses in more natural and meaningful ways.

This talk introduces emotion-aware interfaces and outlines their potential impact across diverse domains, including human-robot collaboration, conversational agents, healthcare, customer engagement, and security. Our research contributions in areas such as facial (micro-)expression recognition, emotional gesture analysis, and remote heart rate estimation from video are presented with the discussion about emerging opportunities and key challenges that must be addressed to fully realize the promise of emotionally intelligent human-machine interaction.

CCS Concepts

• Human-centered computing ~ Human computer interaction (HCI) • Computing methodologies ~ Machine learning

Keywords

Affective computing; Emotion recognition; Emotion-aware interfaces; Multimodal interaction

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Biography

Guoying Zhao received the Ph.D. degree in computer science from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2005. She is currently an Academy Professor and full Professor with University of Oulu and a visiting professor with Aalto University. She is Fellow of IEEE, IAPR and ELLIS, a member of Academia Europaea and member of Finnish Academy of Sciences and Letters. Her



publications have attracted 35100+ citations in Google Scholar with h-index 91. She is Associate Editor-in-Chief for Computer Vision and Image Understanding (CVIU), was/is associate editor for IEEE Trans. on Multimedia, Pattern Recognition, IEEE Trans. on Circuits and Systems for Video Technology, Image and Vision Computing and Frontiers in Psychology Journals. She is/was program co-chair for ECCV 2028 and ICMI 2021, general co-chair for ACII 2025, tutorial chair for ICPR 2024, panel chair for FG 2023, and publicity chair for SCIA 2023 and FG 2018. Her current research interests include affective computing, computer vision, machine learning, and biometrics. Her research has been reported by Finnish TV programs, newspapers and MIT Technology Review.

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